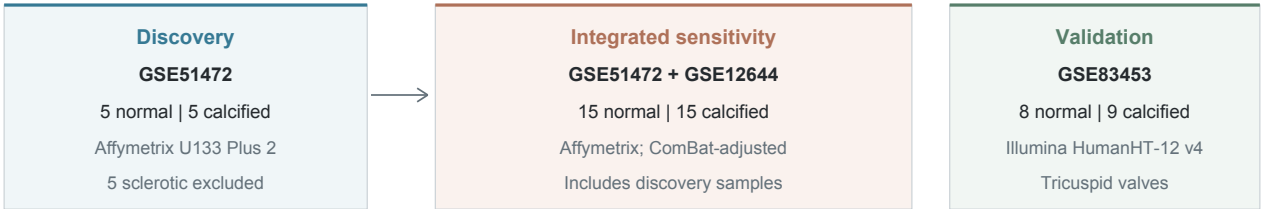
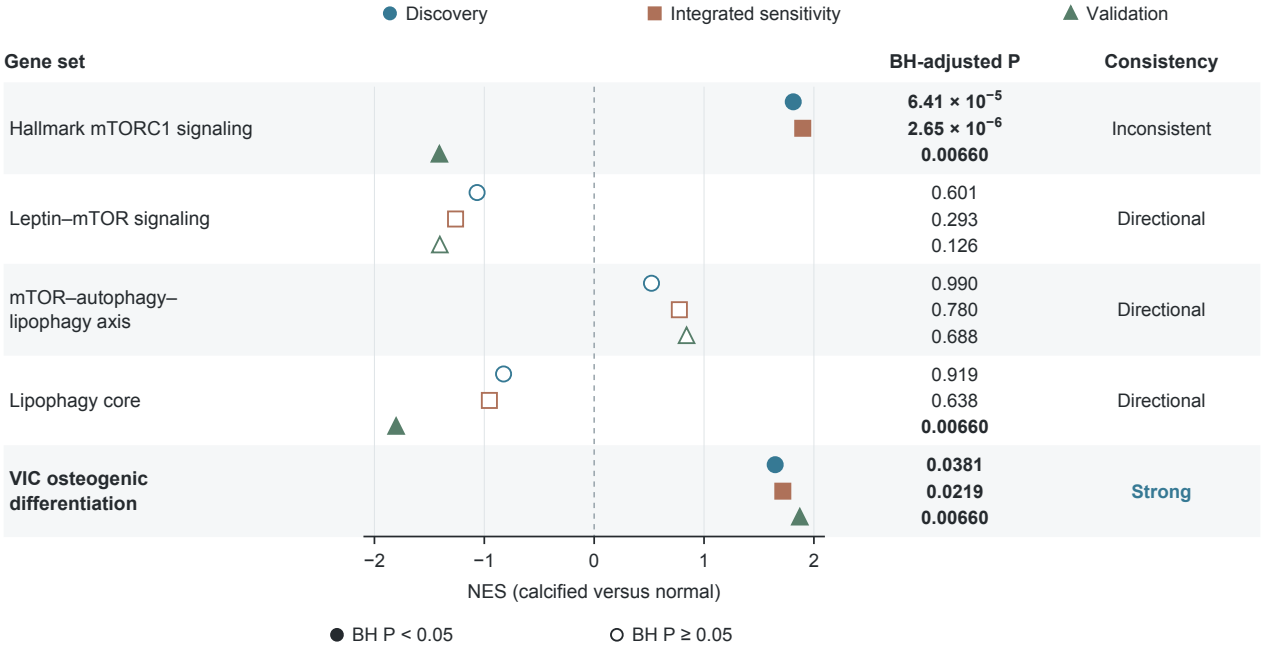


a Human valve transcriptomes



b Prespecified gene-set enrichment



c Osteogenic leading-edge genes

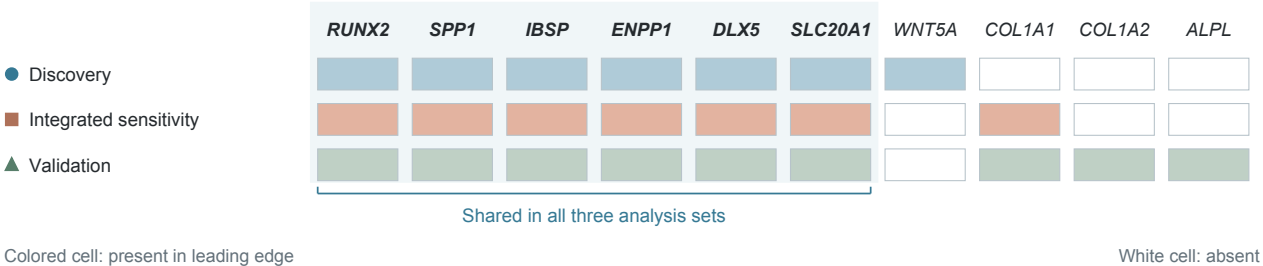


Figure 1. Cross-dataset GSEA identifies consistent enrichment of the VIC osteogenic program. **a** Human valve transcriptomic datasets: GSE51472 (discovery; 5 normal and 5 calcified valves), GSE51472 + GSE12644 (ComBat-adjusted sensitivity analysis; 15 normal and 15 calcified valves), and GSE83453 (validation; 8 normal and 9 calcified tricuspid valves). The sensitivity analysis includes the discovery samples. Sclerotic valves in GSE51472 and bicuspid valves in GSE83453 were excluded. **b** Normalized enrichment scores (NES) for five prespecified gene sets. Positive NES indicates enrichment in calcified valves. Symbols distinguish analysis sets; filled symbols indicate Benjamini–Hochberg-adjusted P < 0.05. Adjustment was performed across the five sets within each analysis set. Strong consistency denotes concordant NES signs with adjusted P < 0.05 in at least two analysis sets; directional consistency denotes concordant signs with fewer than two significant analysis sets; inconsistent denotes opposing signs. **c** Membership of the ten genes in the union of osteogenic leading-edge subsets. Six genes were shared across all three subsets. Exact results and gene membership are provided in Additional file 3: Tables S1 and S2.